

# LECTURE 2: SETS AND FUNCTIONS

Juan Dodyk

WashU

# PLAN

## Sets

1. Extension and Comprehension
2. Operations
3. Intervals

## Functions

1. Domain and Range
2. Inverse functions

GOOD MORNING!



# SETS

A **set** is a collection of things.

If  $A$  is a set and it contains  $x$ , we say that  $x \in A$ . We say that  $x$  is an **element** of  $A$ .

Two sets  $A$  and  $B$  are the same (i.e.,  $A = B$ ) if and only if they have the same elements. In other words, if for every  $x$ ,  $x \in A$  if and only if  $x \in B$ .

We can define a set by extension, i.e., by listing its elements, e.g.,

$$A = \{2, \pi, \text{Juan}\}$$

is the set that contains the numbers 2 and  $\pi$ , and it also contains me.

The **empty set** contains nothing:  $\emptyset = \{\}$ .

Sets can contain other sets, e.g.,  $\{\emptyset, \{\emptyset\}\}$  is a set. In fact, in modern math (with the Zermelo-Fraenkel axioms) everything is a set, so there is nothing else to contain.

# SET COMPREHENSION

We can define a set by comprehension: if  $A$  is a set, we can define the set

$$\{x \in A : P(x)\} \quad \text{or, equivalently,} \quad \{x : x \in A, P(x)\}$$

where  $P(x)$  is a property. For example,

$$\{n \in \mathbb{N} : n < 10 \text{ and } n \text{ is odd}\} = \{1, 3, 5, 7, 9\}.$$

If  $A = \{2, \pi, \text{Juan}\}$ ,

$$\{x \in A : x \text{ is a person}\} = \{\text{Juan}\}.$$

Sometimes people write  $\{x \in A \mid P(x)\}$ .

# OPERATIONS ON SETS

If  $A$  and  $B$  are sets we can define their **union**,  $A \cup B$ , which is the set that contains the elements of  $A$  and those of  $B$ .

We can define their **intersection** as

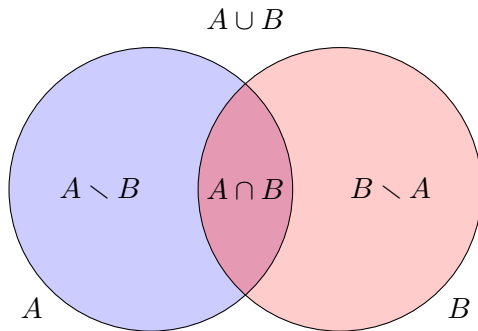
$$A \cap B = \{x \in A : x \in B\},$$

and their **difference** as

$$A \setminus B = \{x \in A : x \notin B\}.$$

We have

- $x \in A \cup B$  if and only if  $x \in A$  **or**  $x \in B$ .
- $x \in A \cap B$  if and only if  $x \in A$  **and**  $x \in B$ .
- $x \in A \setminus B$  if and only if  $x \in A$  **and**  $x \notin B$ .



# NUMBER INTERVALS

If  $a, b \in \mathbb{R}$  we define the **closed interval**

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\},$$

the **open interval**

$$(a, b) = \{x \in \mathbb{R} : a < x < b\},$$

and the interval

$$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}.$$

Also  $[a, +\infty) = \{x \in \mathbb{R} : a \leq x\}$ ,  $(-\infty, a) = \{x \in \mathbb{R} : x < a\}$  and so on.

## QUESTION 1

Write the following sets as intervals:

1.  $[0, 2] \cap [1, 3]$ ,
2.  $[0, 2] \cup [1, 3)$ ,
3.  $[0, 2] \setminus [1, 3]$ .

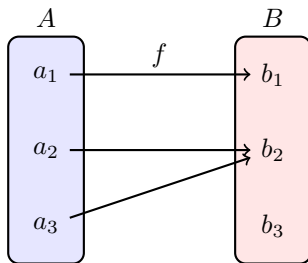
# FUNCTIONS

If  $A$  and  $B$  are two sets a **function**  $f : A \rightarrow B$  is a mapping of each element of  $A$  to an element of  $B$ . If  $x \in A$ ,  $f$  maps it to  $f(x) \in B$ .

We call  $A$  the **domain** of  $f$ , and  $B$  its **codomain**. The **range** of  $f$  is

$$f(A) = \{f(x) : x \in A\}.$$

Example:



Here we have  $f(a_1) = b_1$ , and  $f(a_2) = f(a_3) = b_2$ . The range is  $f(A) = \{b_1, b_2\}$ .



## QUESTION 2

Suppose that  $f : D \rightarrow \mathbb{R}$  is a function that satisfies the following equation for all  $x \in D$ . Find the maximal domain  $D$ .

1.  $f(x) = \frac{1}{x},$
2.  $f(x) = \frac{2x - 1}{\log(x - 1)},$
3.  $f(x) = \sqrt{\exp(x) - 1},$
4.  $f(x) = \log(x) + \log(1 - x).$

# SOME FUNCTIONS

We saw the functions  $\exp : \mathbb{R} \rightarrow (0, +\infty)$  and  $\log : (0, +\infty) \rightarrow \mathbb{R}$ .

**Linear** functions are  $f : \mathbb{R} \rightarrow \mathbb{R}$  given by  $f(x) = ax + b$  for some  $a, b \in \mathbb{R}$ .

**Quadratic** functions are  $f : \mathbb{R} \rightarrow \mathbb{R}$  given by  $f(x) = ax^2 + bx + c$  for some  $a, b, c \in \mathbb{R}$  with  $a \neq 0$ .

**Polynomial** functions are  $f : \mathbb{R} \rightarrow \mathbb{R}$  given by  $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$  for some  $n \in \mathbb{N}$  and  $a_0, \dots, a_n \in \mathbb{R}$ .

Example:  $f(x) = 2x^5 - 3x^2 + 1$ .

# COMPLETING THE SQUARE

If  $x, y \in \mathbb{R}$  we have  $\boxed{(x \pm y)^2 = x^2 \pm 2xy + y^2}$ .

In particular,  $\left(x \pm \frac{1}{2}b\right)^2 = x^2 \pm bx + \left(\frac{b}{2}\right)^2$ .

Therefore,  $x^2 \pm bx = \left(x \pm \frac{1}{2}b\right)^2 - \frac{1}{4}b^2$ .

Consider  $x^2 - x + 1$ .

$$x^2 - x = \left(x - \frac{1}{2}\right)^2 - \frac{1}{4}.$$

$$\text{So } x^2 - x + 1 = \left(x - \frac{1}{2}\right)^2 - \frac{1}{4} + 1 = \left(x - \frac{1}{2}\right)^2 + \frac{3}{4}.$$

## QUESTION 3

What is the range of  $f(x) = x^2 - x + 1$ ?

# QUADRATIC FUNCTIONS: ROOTS

A **root** of a function  $f$  is a point  $x$  such that  $f(x) = 0$ .

How to find the roots of a quadratic? We can complete the square. For example, to solve

$$x^2 - 4x + 1 = 0$$

we complete the square first:  $x^2 - 4x + 1 = (x - 2)^2 - 4 + 1 = (x - 2)^2 - 3$ .

We want  $(x - 2)^2 - 3 = 0$ .

This is  $(x - 2)^2 = 3$ .

Two options:  $x - 2 = \sqrt{3}$  or  $x - 2 = -\sqrt{3}$ .

So, two roots:  $x = 2 + \sqrt{3}$  and  $x = 2 - \sqrt{3}$ .

## QUESTION 4

What about  $f(x) = x^2 - x + 1$ ?

# QUADRATIC FUNCTIONS: ROOTS

There is a formula: if  $f(x) = ax^2 + bx + c$  with  $a \neq 0$  the roots are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

This only works if  $b^2 - 4ac \geq 0$ .

If  $b^2 - 4ac < 0$  then the quadratic doesn't have roots.

## QUESTION 5

Use the formula for  $x^2 - x + 1 = 0$  and  $x^2 - 4x + 1 = 0$ .

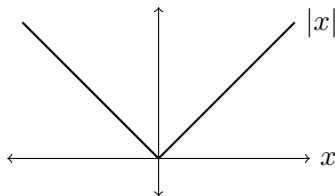
# ABSOLUTE VALUE

We define the **absolute value** function  $|\cdot| : \mathbb{R} \rightarrow [0, +\infty)$  by

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Example:  $|2| = 2$ ,  $|0| = 0$ , but  $|-1| = 1$ .

We have  $|xy| = |x||y|$  and  $|x + y| \leq |x| + |y|$  for all  $x, y \in \mathbb{R}$ .



# INVERSE FUNCTIONS

Let  $f : X \rightarrow Y$  be a function. Suppose that

for each  $y \in Y$  there is exactly one  $x \in X$  such that  $f(x) = y$ .

In that case we say that  $f$  is **invertible** or a **bijection**.

We obtain a function  $f^{-1} : Y \rightarrow X$  such that  $f(f^{-1}(y)) = y$  for all  $y \in Y$ . We call it the **inverse** function of  $f$ .

# INVERSE FUNCTIONS

“ $f^{-1}$  is the inverse of  $f$ ” is equivalent to

for every  $x \in X$ ,  $y \in Y$ ,  $f(x) = y$  if and only if  $f^{-1}(y) = x$ .

Notice the symmetry: if  $f^{-1}$  is the inverse of  $f$  then  $f$  is the inverse of  $f^{-1}$ .

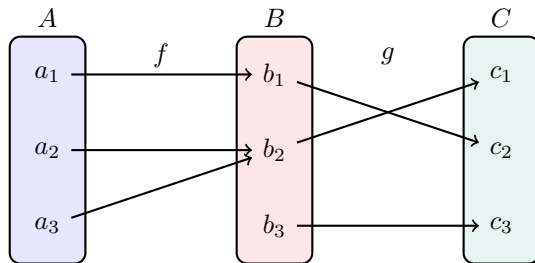
## Examples.

1. If  $f(x) = x + 1$  then  $f^{-1}(x) = x - 1$ , since  $x + 1 = y$  if and only if  $x = y - 1$ .
2. The inverse of  $f(x) = x^{\frac{1}{2}}$  is  $g(x) = x^2$  defined only for  $x \geq 0$ .
3.  $\log$  is the inverse of  $\exp$ .
4.  $\log_a(x)$  is the inverse of  $f(x) = a^x$  if  $a > 0$ ,  $a \neq 1$ .
5. The function  $f(x) = 1/x$  is its own inverse.
6. The absolute value  $|\cdot|$  doesn't have an inverse.

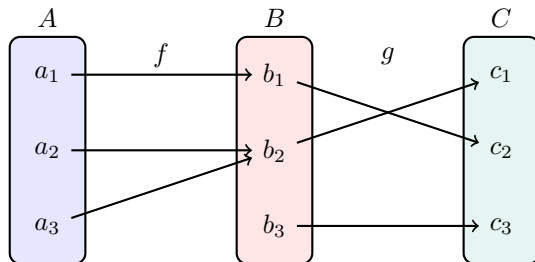


### QUESTION 6

Do either  $f$  or  $g$  have an inverse? If so, find it.



# ANSWER



| $f$   |       |
|-------|-------|
| $x$   | $y$   |
| $a_1$ | $b_1$ |
| $a_2$ | $b_2$ |
| $a_3$ | $b_2$ |

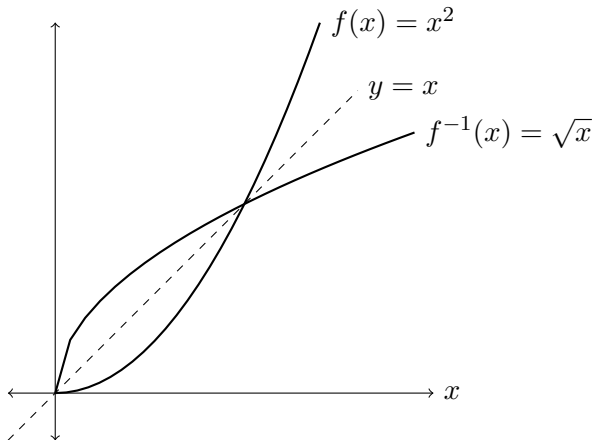
| $f^{-1}$ doesn't exist |                  |
|------------------------|------------------|
| $x$                    | $y$              |
| $b_1$                  | $a_1$            |
| $b_2$                  | $a_2$ or $a_3$ ? |
| $b_3$                  | ?                |

| $g$   |       |
|-------|-------|
| $x$   | $y$   |
| $b_1$ | $c_2$ |
| $b_2$ | $c_1$ |
| $b_3$ | $c_3$ |

| $g^{-1}$ |       |
|----------|-------|
| $x$      | $y$   |
| $c_1$    | $b_2$ |
| $c_2$    | $b_1$ |
| $c_3$    | $b_3$ |

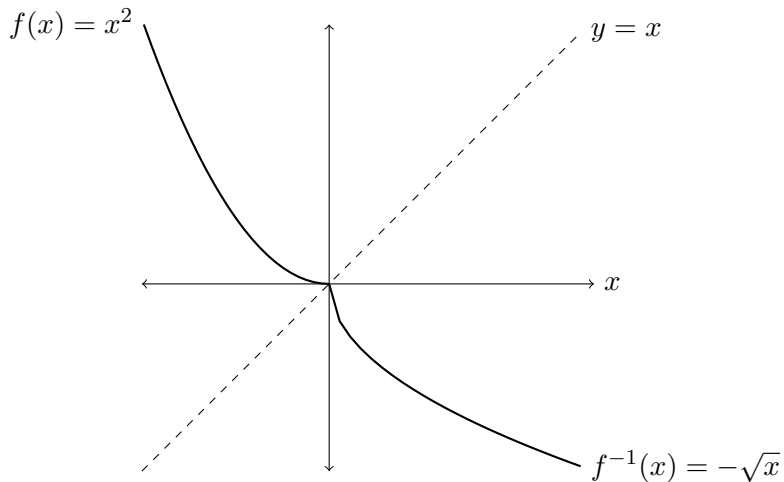
# INVERSES AS REFLECTIONS ABOUT THE $45^\circ$ LINE

$$f : [0, +\infty) \rightarrow [0, +\infty), f(x) = x^2$$



# INVERSES AS REFLECTIONS ABOUT THE $45^\circ$ LINE

$$f : (-\infty, 0] \rightarrow [0, +\infty), f(x) = x^2$$



### QUESTION 7

Does  $f$  have an inverse? If so, find it.

1.  $f : \mathbb{R} \rightarrow \mathbb{R}$  given by  $f(x) = 2x - 1$ ,
2.  $f : \mathbb{R} \rightarrow [1, +\infty)$  given by  $f(x) = x^2 + 1$ ,
3.  $f : [0, +\infty) \rightarrow [1, +\infty)$  given by  $f(x) = x^2 + 1$ ,
4.  $f : (-\infty, 0] \rightarrow [1, +\infty)$  given by  $f(x) = x^2 + 1$ .

